

2.1 Sequential Models and RNNs

A hidden state that reads the sequence one token at a time, with shared weights.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

An MLP sees a vector and forgets. Language is a **sequence**: the meaning of *not* depends on what came before. A recurrent net keeps a **state** and updates it at every token.

This week is the historical path to transformers. You still need it: LSTMs show up in speech and time series, and the failure mode (a chain of Jacobians) is why Week 3 exists.

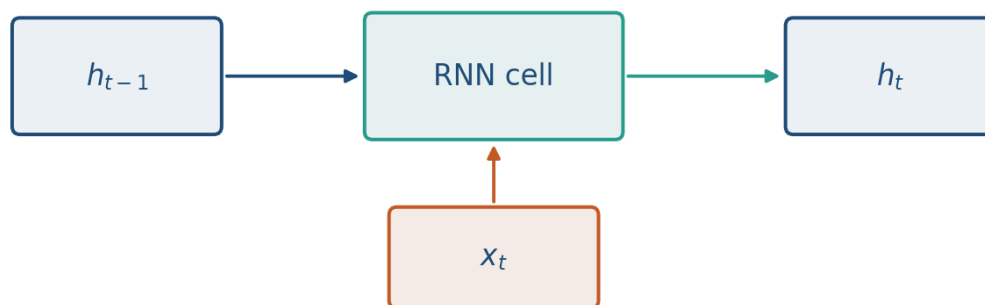
1. The cell

At step t the cell takes

- the new token embedding x_t , and
- the previous hidden state h_{t-1} .

It emits h_t , which is supposed to hold **both**. That vector is the memory.

Same weights at every step: input plus previous state



The same weights are used at every step. That is weight sharing along time, the sequence analogue of a CNN's shared filter.

Between two independent sequences (two reviews in a batch) you **reset** $h_0 = 0$. The loop lives *inside* one sequence.

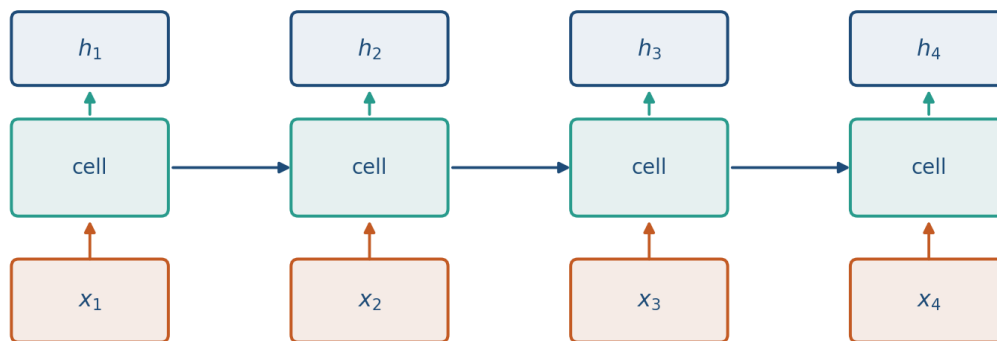
If you do not reset, review 2 starts with leftover sentiment from review 1. That is leakage, not transfer learning.

2. Unrolled in time

Write the loop as a chain. Three tokens, one set of weights:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b).$$

Unrolled in time. The chain is why long-range gradients struggle.



A CNN looks at a **fixed** neighborhood, in parallel. An RNN looks at **everything so far**, in order, and cannot fully parallelize across time. That is the accuracy / speed tradeoff.

Unrolling is how you backprop. The picture is one graph with copies of W_h, W_x . There are not three different matrices named $W_h^{(1)}, W_h^{(2)}, W_h^{(3)}$.

3. What the state has to carry

If the first word is the subject of a verb twenty tokens later, h_t must still contain that subject. The cell is a lossy compressor: every step overwrites. Week **2.2** is what happens to the gradient on that long path. Week **2.3** adds gates so the cell can choose what to keep.

A many-to-one RNN (sentiment) uses the last h_T . A many-to-many RNN (tagging, language modeling) emits at every t . Next-token LM: predict x_{t+1} from h_t .

4. Teaching this note

30–40 minutes. Draw the cell, unroll three tokens with **shared** W_h , then do the 2-D numeric example with $h_0 = 0$. Play **0:00–12:00** of StatQuest RNNs (the loop and the unroll). Save vanishing gradients for note 2.2 even if the video teases them.

5. Worked example

Let $d = 2, b = 0, h_0 = 0$,

$$W_h = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}, \quad W_x = I, \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Step 1: $W_h h_0 + W_x x_1 = x_1$, so $h_1 = \tanh(x_1) \approx \begin{bmatrix} 0.76 \\ 0 \end{bmatrix}$ because $\tanh(1) \approx 0.76$.

Step 2: $W_h h_1 + W_x x_2 \approx \begin{bmatrix} 0.38 \\ 1 \end{bmatrix}$, then $h_2 = \tanh(\cdot) \approx \begin{bmatrix} 0.36 \\ 0.76 \end{bmatrix}$.

The same W_h was used twice. Token 1 still has a **shrunk** footprint in h_2 ($0.76 \rightarrow 0.36$). That shrinkage is the teaser for vanishing gradients.

6. Where students get stuck

- Drawing a new W_h at each time step (no sharing).
 - Feeding the whole sentence as one concatenated vector “to make it an MLP.”
 - Forgetting to reset h between batch items.
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7. Video

Watch [StatQuest: Recurrent Neural Networks, Clearly Explained](#).

Pause on the first unroll (same weights copied) and on the moment the hidden state is described as memory. Skip any long software demo; the board math is the point.

8. Practice

1. Unroll three tokens on paper with $d = 2$, $h_0 = 0$. Which matrices are reused?
2. Why reset the state between two unrelated documents in a batch?
3. Name one thing an RNN can mix that a kernel of size 5 cannot, in a single layer.
4. Using the worked example's W_h, W_x and $x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, write the pre-activation $W_h h_2 + W_x x_3$ with $h_2 \approx \begin{bmatrix} 0.36 \\ 0.76 \end{bmatrix}$. Do not bother with $\tanh$ if you are short on time; the linear mix is the lesson.
5. If every W_h were replaced by $0.1I$, and $x_t = 0$ after $t = 1$, approximately how does $\|h_t\|$ scale with t ? (Geometric sequence.)